SDP and Quadratic Optimization

$$\min: \quad x^T Q_0 x + q_0^T x + c_0$$

$$\text{st}: \quad x^T Q_i x + q_i^T x + c_i \leq 0, \quad i = 1, \ldots, m$$

Need for convexity:

$$Q_0 \succeq 0$$

$$Q_i \succeq 0$$

* Note: if we had

$$x^T Q_i x + q_i^T x + c_i \geq 0$$

This constraint is convex if $Q_i \preceq 0$. *

Preliminaries

$$Q \succeq 0 \quad \Longleftarrow \quad Q = M^T M \quad \text{or} \quad MM^T$$

Recall: $x^T Q x \geq 0 \quad \forall x.$ If $Q = M^T M$

$$\Rightarrow x^T Q x = \|Mx\|_2^2$$

Now we show $\Rightarrow$

$$Q = \sum \lambda_i v_i v_i^T \quad . \qquad \text{Since } Q \succeq 0 \quad \lambda_i \geq 0$$

$$M = \sum \sqrt{\lambda_i}\, v_i v_i^T$$

Claim: $M^T M = Q$. $\quad \left(\sum_i \sqrt{\lambda_i}\, v_i v_i^T\right)\left(\sum_j \sqrt{\lambda_j}\, v_j v_j^T\right) = \sum_{i,j} \sqrt{\lambda_i}\sqrt{\lambda_j}\, v_i v_i^T v_j v_j^T$

$$= \sum_i \lambda_i v_i v_i^T = Q.$$

$$X = \begin{pmatrix} A & B \\ B^T & C \end{pmatrix}$$

$$X \succeq 0 \iff A \succeq 0 \quad C - B^T A^{-1} B \succeq 0$$

Reformulating QCQP as an SDP — 1/3

min: $x^T Q_0 x + q_0^T x + c_0$

s.t.: $x^T Q_i x + q_i^T x + c_i \leq 0$

$$\left(\begin{array}{c|c} I & M_i x \\ \hline x^T M_i^T & -c_i - q_i^T x \end{array}\right) \succeq 0 \iff x^T Q_i x + q_i^T x + c_i \leq 0$$

where $Q_i = M_i^T M_i$

$A \succ 0, \quad C - B^T A^{-1} B \succeq 0$

SC: $$\begin{array}{c} \begin{array}{cc} A & B \end{array} \\ \left(\begin{array}{c|c} I & M_i x \\ \hline x^T M_i^T & -c_i - q_i^T x \end{array}\right) \\ B \quad\quad C \end{array} \succeq 0 \iff -c_i - q_i^T x - x^T \underbrace{M_i^T M_i}_{Q_i} x \geq 0$$

constraints.

Objective:

min: θ

st: $x^T Q_0 x + q_0^T x + (c_0 - \theta) \leq 0$

$\vdots$

Reformulating QCQP as an SDP — 2/3

$$\min: \; x^T Q_0 x + q_0^T x + c_0$$
$$\text{st.} \; x^T Q_i x + q_i^T x + c_i \le 0, \quad i = 1, \dots, m$$

$$\Updownarrow$$

$$\min: \; \theta$$
$$\text{st:} \; \begin{pmatrix} I & M_0 x \\ x^T M_0 & -c_0 - q_0^T x + \theta \end{pmatrix} \succeq 0$$

$$\begin{pmatrix} I & M_i x \\ x^T M_i & -c_i - q_i^T x \end{pmatrix} \succeq 0, \quad i = 1, \dots, m.$$

Reformulating QCQP as an SDP — 3/3

An alternate reformulation.

$$\begin{pmatrix} 1 & x^T \\ x & W \end{pmatrix} \succeq 0 \iff W \succeq xx^T \qquad (W - xx^T) \succeq 0.$$

Check: follows from Schur.

Next:

$$\left\langle \begin{pmatrix} c_i & \frac{1}{2} q_i^T \\ \frac{1}{2} q_i & Q_i \end{pmatrix}, \begin{pmatrix} 1 & x^T \\ x & W \end{pmatrix} \right\rangle \leq 0 \qquad \& \begin{pmatrix} 1 & x^T \\ x & W \end{pmatrix} \succeq 0$$

$$\iff \boxed{x^T Q_i x + q_i^T x + c_i \leq 0. \quad (*)}$$

$$\langle A, B \rangle = \sum_{i,j} A_{ij} B_{ij} \quad // \quad c_i + q_i^T x + \underline{\langle Q_i, W \rangle} \leq 0$$

$$Q_i \succeq 0 \Rightarrow \langle Q_i, W \rangle \geq \langle Q_i, xx^T \rangle = x^T Q_i x$$

Putting this together, we have:

$$\min: \ \theta$$

$$\text{st}: \ \left\langle \begin{pmatrix} c_0 - \theta & \frac{1}{2} q_0^T \\ \frac{1}{2} q_0 & Q_0 \end{pmatrix}, \begin{pmatrix} 1 & x^T \\ x & W \end{pmatrix} \right\rangle \leq 0$$

ordinary inequality

$$\left\langle \begin{pmatrix} c_i & \frac{1}{2} q_i^T \\ \frac{1}{2} q_i & Q_i \end{pmatrix}, \begin{pmatrix} 1 & x^T \\ x & W \end{pmatrix} \right\rangle \leq 0, \quad i = 1, \dots, m$$

$$\begin{pmatrix} 1 & x^T \\ x & W \end{pmatrix} \succeq 0$$

$m+1$ linear constraints.

We have more variables: x n

W $\frac{n \cdot n+1}{2}$

However: only 1 semidef const.